

Collaborative Development of New
Strategic Planning Models – Pooled Fund
Phase 2

ITHIM Integration Task 8.2

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VE-ITHIM Integration

The goal of Task 8 of the VisionEval (VE) Pooled Fund project is to demonstrate how the VE system can be integrated with ITHIM, one of the leading transportation health impact modeling tools today. The first deliverable within Task 8 developed design recommendations for this integration. This document described the resulting prototype integration. The prototype is referred to as VE-ITHIM and is a standalone set of scripts to work for a specific VE-RSPM model geography to create ITHIM input files and then uses direct model results from a full VE-RSPM run in a sequence of R scripts to run ITHIM procedures. VE-ITHIM is fully independent and can be appended to any VE-RSPM model and does not affect the core VE-RSPM framework or run scripts.

Overview of ITHIM

The Integrated Transport and Health Impact Modeling Tool (ITHIM) was originally developed as a spreadsheet model in 2009 (Woodcock et al.) Subsequent versions of the model have been implemented in Analytica (ITHIM 2.0) and R (ITHIM-R). A handful of ITHIM models have been developed for cities across the United States, including San Francisco, CA,¹ Nashville, TN,² and Sacramento, CA.³ Newer US-based implementations of ITHIM have used the R version of the model (ITHIM-R).

Integrating ITHIM and VE

The project team developed a set of requirements to demonstrate how ITHIM can be applied to estimate the health impacts of VE scenarios. These requirements, which guided the development the prototype ITHIM model, are summarized below:

1. Derive a synthetic population and trip set from the VE household table.
2. Focus on five key health outcomes (all-cause mortality, diabetes, lung cancer, chronic obstructive pulmonary disease, and road injuries)
3. Maintain geographic disaggregation throughout model runtime.

¹ Maizlish, N., Woodcock, J., Co, S., Ostro, B., Fanai, A., & Fairley, D. (2013). Health cobenefits and transportation-related reductions in greenhouse gas emissions in the San Francisco Bay area. *American journal of public health*, 103(4), 703-709.

² Whitfield, G. P., Meehan, L. A., Maizlish, N., & Wendel, A. M. (2017). The integrated transport and health impact modeling tool in Nashville, Tennessee, USA: implementation steps and lessons learned. *Journal of transport & health*, 5, 172-181.

³ Karner, A., Rowangould, D., Wu, Y., Igbinedion, O., London, J. (2017). Development and Application of an Integrated Health Impacts Assessment Tool for the Sacramento Region. A Research Report from the National Center for Sustainable Transportation. University of California, Davis.

4. Remove features from ITHIM-R that are optional and/or duplicate existing VE functionality.
5. Improve settings/parameter management.

Choosing a “Base” ITHIM Model for Integration with VE

Because multiple implementations of ITHIM exist today, an important consideration in integrating ITHIM with the VE framework was which implementation will serve as the base ITHIM model. The project team identified two candidates: the officially maintained core ITHIM-R model and the California ITHIM model. The California ITHIM model is simpler, but less modular than ITHIM-R while ITHIM-R contains more detailed implementations of exposure pathways, in particular the air pollution and traffic safety pathways. To leverage the more modular design of ITHIM-R, the project team decided to use this implementation of the model as a basis for the air pollution and physical activity pathways. However, ITHIM-R includes a complicated traffic safety pathway (including the estimation of crash prediction regression models on-the-fly) that presented substantial barriers to implementation in the prototype model discussed here. Thus, the project team adopted the simpler rates-based approach to estimate crashes implemented in the California ITHIM model for the prototype model.

ITHIM-R Model Structure

ITHIM-R is organized into a series of modules which are executed by a runner script *run_ithim.R*. Model configuration and parameters are managed by an *ithim_object* and/or stored in the global environment. Different exposure scenarios (including the baseline scenario) are represented via different sets of trips for members of a synthetic population; these exposure scenarios are used to calculate relative risks and health burdens relative to the baseline scenario for each pathway. All other model parameters are held constant across scenarios. ITHIM-R can be executed in batch mode to explore multiple sub-geographies; however, this is accomplished by looping over the functions described below rather than through vectorization which limits the scalability of the model to large numbers of sub-geographies.

The core functionality of ITHIM-R is distributed across ten core modules, which are called sequentially to evaluate the health impacts of a scenario. Given at least two sets of trips (a reference scenario and at least one scenario), the air pollution pathway is evaluated first using three modules (*scenario_pm_calculations*, *scenario_co2_calculations*, and *gen_ap_rr*). Next, the physical activity pathway is evaluated using two modules (*total_mmet* and *gen_pa_rr*), and the combined health burden of air pollution and physical activity are evaluated using the *combined_rr_ap_pa* and *health_burden* modules. The traffic safety pathway is evaluated by the *injuries_function_2* and *injury_death_to_yll* modules and the cumulative health burden is estimated using the *join_hb_and_injury* module.

ITHIM Data Requirements

ITHIM is a data-intensive model. Input data requirements include a synthetic population with age and sex characteristics, daily minutes spent walking and biking for transportation for members of the synthetic population, distance travelled by members of the synthetic population by travel mode, non-transportation physical activity, and other inputs. Broadly, input data requirements can be grouped into five categories:

- **Population data.** Data describing the synthetic population for which health outcomes are modeled.
- **Trip data.** Data describing travel behavior of the population (i.e., trip distances and durations by mode).
- **Baseline health status data.** Data characterizing the baseline health status of the population.
- **Baseline exposure data.** Data capturing exposures and/or behaviors that impact human health, both in the baseline and in any modelled scenarios.
- **Epidemiological/health risk data.** Parameters that establish a relationship between an exposure and one or more health outcomes.

In the Design Recommendations document, ITHIM data requirements were mapped to VE inputs and outputs. Critical linkages to the VE framework include population and trip data, while other ITHIM data requirements are met through third-party data sources or by adopting parameters in the ITHIM-R source code.

Prototype Model Design

The prototype VE/ITHIM model is organized into five subcomponents. First, VE outputs are transformed to align data schemas with those expected by downstream ITHIM functions (Subcomponent 1: Data Harmonization). Next, the air pollution pathway is evaluated using refactored functions from the ITHIM-R source code (Subcomponent 2: Air Pollution Pathway). The physical activity pathway is then evaluated, again using refactored functions from the ITHIM-R source code (Subcomponent 3: Physical Activity Pathway). The traffic safety pathway is then evaluated using functions that implement an approach like the rates-based approach in California ITHIM (Subcomponent 4: Traffic Safety Pathway). Finally, the cumulative health burden is calculated using functions re-factored from the ITHIM-R source code (Subcomponent 5: Health Burden). These subcomponents are looped over for each scenario and evaluation year and executed via a runner script.

The project team also worked to remove some features from the model to simplify this initial instance. Core ITHIM functions that calculate CO₂ emissions were removed because this functionality is duplicative of existing VE features. Additionally, some uncertainty features (e.g., using upper- and lower-bound relative risk values when calculating health burdens) were removed in the prototype. These features may be re-introduced in the future if desired. Finally, the project

team improved settings management by reducing reliance on the global environment and reducing the number of hard-coded parameters in the code. Further refinements can be made to settings management, but the prototype model represents a substantive improvement over the approach in ITHIM-R. The LoadSettings.R module is called early in model execution and contains a single function to load settings into a single object that can then be passed to downstream functions.

The five model subcomponents listed above are described in greater detail below.

Subcomponent 1: Data Harmonization

This subcomponent includes two modules: HarmonizeData.R and GenerateScenarios.R. The first module, HarmonizeData.R contains functions that decompose VE households to persons, estimate ventilation rates for individuals in the person table, and join zone-based background exposures (PM_{2.5} concentrations and leisure-time physical activity estimates) to the person table. While some of this functionality is new, the approach to estimating ventilation rates (*estimateVentilationRates* function) is taken from the ITHIM-R source code but placed further upstream.

The second module, GenerateScenarios.R, contains functions that append VE travel estimates from the counterfactual scenario being assessed to the persons table and converts the persons table to a *pseudo* trips table. Establishing this *pseudo* trip table, hereafter referred to as the trip table, is an important step in integrating downstream functions as ITHIM-R operates largely on a trips dataset when evaluating exposures.

Input data: VE household output csv files, bzone PM_{2.5} and leisure-time physical activity estimates, and ventilation rate calculation parameters.

Output data: Person table, trips table.

Subcomponent 2: Air Pollution Pathway

The air pollution pathway subcomponent includes two modules: ITHIMCoreAP.R and APRiskFuncs.R. The first module contains minimally re-factored versions of functions from the ITHIM-R source code that estimate total PM_{2.5} exposure for individuals in the person table, accounting for trips taken by those individuals, leisure-time physical activity, and assumptions about other activities undertaken during a typical day (e.g., inhalation while sleeping).

The second module in the air pollution subcomponent translates PM_{2.5} exposure estimates into risk values and calculates the difference in risk between the scenario being evaluated and the reference scenario. This risk difference is used in downstream health burden functions to estimate health impacts associated with PM_{2.5} exposure.

Input data: Person table, trips table, PM_{2.5} exposure parameters.

Output data: PM exposure relative risk table.

Subcomponent 3: Physical Activity Pathway

The third subcomponent contains two modules: ITHIMCorePA.R and PARiskFuncs.R. Like the previous subcomponent, these modules contain minimally re-factored functions from the ITHIM-R source code that estimate total physical activity levels for each scenario and the risk values associated with these activity levels.

Input data: Person table, trips table, PM_{2.5} exposure parameters.

Output data: PA exposure relative risk table.

Subcomponent 4: Traffic Safety Pathway

The traffic safety subcomponent contains a single module, Crashes.R. This module calculates crash rates, accounting for the travel mode of both the victim and the striking vehicle and applies these rates to estimate crashes for each scenario. Unlike the previous subcomponents, the traffic safety pathway adopts the approach used in California ITHIM. This is largely because the traffic safety pathway in ITHIM-R uses a probabilistic approach designed to work with more detailed representation of travel modes than is produced by VE (i.e., including motorcycle crashes and other more specialized modes like rickshaws). The method implemented here is a mechanistic, rates-based approach that better integrates with VE mode choice set (auto, transit, walk, and bike). The traffic safety pathway can use observed crash data for the region or, if available, rates from the *marea_safety_factors.csv* file used for the VE Travel Performance package. The desired crash input approach can be specified in the *config.csv* input file described below.

Input data: Trips table, input crash data (FARS data or *marea_safety_factors*).

Output data: Injury predictions table.

Subcomponent 5: Health Burden

The final subcomponent contains a single module, HealthBurden.R, that estimates the combined health burden across the three modeled exposure pathways (air pollution, physical activity, and traffic safety). Health outcomes include both mortality and years of life lost (YLL). The functions in this module are minimally re-factored versions of functions from the ITHIM-R source code.

Input data: PM exposure relative risk table, PA exposure relative risk table, injury predictions table, persons table, input burden of disease data.

Output data: Health burden tables.

Input Data Requirements

The prototype module introduces several new data requirements. Some of these files will need to be updated for each VE-ITHIM application while other files provide underlying ITHIM data and/assumptions and should not be changed on a project-by-project basis. The first group of files—those that need to be updated for each application—are expected to be the module's data directory (i.e., */inst/extdata*). These files are listed below, along with source data for each:

- **config.csv:** this file contains configuration parameters for VE-ITHIM. These values must be updated for each new VE-ITHIM application and are listed below:
 - **model_name:** the name of the VE model (e.g., VERSPM-scenarios)
 - **base_model:** the name of the VE base scenario (e.g., stage-base)
 - **base_year:** the year of the VE base scenario, (e.g., 2021)
 - **reference_model:** the name of the VE reference scenario (e.g., stage-future)
 - **crash_method:** must be one of “ve” or “fars”; defines which input data file to use when modeling the traffic safety pathway (*fars.csv* or *marea_safety_factors.csv*)
- **settings.csv:** this file contains several parameters that are required to run ITHIM modules, with default values adopted from the existing ITHIM-R source code. Default parameter values can be updated using local data if available. Source: ITHIM-R source code.
- **bzone_pm.csv:** this file contains estimates of background PM_{2.5} concentrations for each bzone in the model region: Source: Atmospheric Composition Analysis Group at Washington University in St. Louis estimates spatially joined to model bzones.⁴ Example code allocating the PM concentration to bzones is provided in */scripts/prepare_pm_inputs.py*.
- **bzone_pa.csv:** this file contains estimates of leisure time physical activity (LTPA) for individuals living in each bzone in the model region, presented as distribution across four physical activity levels (no activity; 40 to 74 minutes per week, 75 to 149 minutes per week, or 150 to 225 minutes per week). Source: the Centers for Disease Control and Prevention (CDC) PLACES dataset⁵ and Behavioral Risk Factor Surveillance System (BRFSS) survey⁶, processed and allocated to model bzones scripts using an approach provided in */scripts/prepare_pa_inputs.py*
- **fars.csv** (*optional, one of fars.csv or marea_safety_factors.csv must exist*): this file contains all transportation fatalities in the study region, pooled over a five-year period, along with the travel mode of the victim and the mode travel mode of the striking vehicle (i.e., records in “who-hit-who” format). Source: Fatality Analysis Reporting System (FARS) data processed and subset to include only families in the region.⁷ Example code allocating the data to zones is provided in */scripts/prepare_fars_inputs.py*

⁴ <https://sites.wustl.edu/acag/datasets/surface-pm2-5/#V5.GL.04>

⁵ <https://www.cdc.gov/places/index.html>

⁶ https://www.cdc.gov/brfss/annual_data/annual_data.htm

⁷ <https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system-fars>

- **marea_safety_factors.csv** (optional, one of *fars.csv* or *marea_safety_factors.csv* must exist): this file is a copy of the *marea_safety_factors.csv* file that may be present on the VE model's input data.
- **gbd.csv**: this file contains age-and sex-specific death and years of life lost (YLL) estimates for each disease present in *disease_outcomes_lookup.csv* (i.e., each modeled health outcome). Any health outcome estimated by the model needs to be present in this file. Source: Institute of Health Metrics and Evaluation (IHME) global burden of disease estimates.⁸
- **disease_outcomes_lookup.csv**: this file contains a crosswalk of all modeled health outcomes, mapping the outcome to the disease name in *gbd.csv* and well as in the relative risk input csvs. Any health outcome estimated by the model needs to be present in this file. Source: ITHIM-R source code.

The second group of input files—those that provide underlying ITHIM data and/or assumptions—are expected to be in the three subfolders (*/ithim*, */drpa*, and */drap*) within the */inst/extdata* directory. The */ithim* directory contains input files that define inhalation rates:

- **BodyMass.csv**: this file contains parameters used to estimate body mass which, along with the next four input files, are required for ventilation rate calculations. Default values have been adopted from the existing ITHIM-R source code. Source: ITHIM-R source code.
- **ECF.csv**: this file contains parameters used to estimate individual's energy conversion factors. Source: ITHIM-R source code.
- **RMR.csv**: this file contains parameters used to estimate resting metabolic rates. Source: ITHIM-R source code.
- **NVO2max.csv**: this file contains parameters used to estimate maximum oxygen uptake rates. Source: ITHIM-R source code.
- **VentilationFromOxygenUptake.csv**: this file contains parameters used to estimate ventilation from oxygen uptake. Source: ITHIM-R source code.

Finally, the */drpa* and */drap* directories contains many files that define relative risks values for various disease outcomes for the air pollution and physical activity health outcome pathways. The contents of these files are not detailed here.

Running the Prototype Model

For testing purposes, the prototype model has been provided alongside this report in a compressed folder, including input data required to run the case study application described below. The prototype model code is also available in the 'ithim' branch of RSG's VisionEval-Extras repository

⁸ <https://www.healthdata.org/data-tools-practices/interactive-visuals/gbd-compare>

on GitHub, which can be transferred to FHWA's repository via a pull request.⁹ The prototype model is implemented as post-process once a set of VE scenarios have been run locally and VE results have been exported to csv files. In its current form, the workflow for running the prototype model includes:

1. **Ensure required R libraries are installed.** VE-ITHIM uses a minimal set of R libraries (data.table, tidyr, pracma, and dplyr). Install these libraries if they have not been installed locally.
2. **Unzip the compressed "VEIthim" folder or clone the VisionEval-Extras repository and switch to the 'ithim' branch.** Because the prototype model is a standalone tool, it does not need to be installed locally, but can be run from source in RStudio.
3. **Export csv files from the VE datastore and copy the VE model to the /models subdirectory.** The prototype models expect a full copy of a VE model, including exported csvs, to be present in the /models subdirectory. An example of this can be found in the zipped VEIthim folder, which contains the case study model described below.
4. **Update VE-ITHIM configuration as required.** The VE-ITHIM configuration file contains details necessary to run the model as detailed in the *Input Data Requirements* section above. Update these fields to be consistent with the VE model contained in the /models subdirectory.
5. **Update data input files as required.** As detailed in the *Input Data Requirements* section above, several Python files are provided in the scripts directory to help develop local inputs. Other inputs require the user to visit the source location to obtain region and model specific parameter values. Running these scripts will write required input files to disk in the correct location (i.e., the /inst/extdata subdirectory).
6. **Run RunITHIM.R.** Once all input data requirements have been met, *RunITHIM.R* will execute the prototype model and write output csvs to disk in the directory containing csvs exported from VE. These output files will have the prefix "ithim_".

Case Study Application of Prototype

To demonstrate the functionality of the prototype model, health impact estimates were developed for four hypothetical scenarios for the Salem, Oregon region (SKATS VE-RSPM), relative to a reference scenario. The VE-RSPM for the Salem region was developed for a separate use and was available to produce outputs for the purposes of testing VE-ITHIM. These scenarios are:

- **Reference:** The VE-RSPM for the SKATS MPO created a 2050 reference scenario aligned with the regional long range transportation plan.

⁹ <https://github.com/RSGInc/VisionEval-Extras/tree/ithim>

- **SOV diversion:** 50% increase in single-occupancy vehicle (SOV) diversion rates. The values in the *azone_prop_sov_dvmt_diverted.csv* input file were increased by 50%.
- **Congestion charge:** addition of a \$0.25 per vehicle mile charge to severe and extreme congestion. The *marea_congestion_charges.csv* file was revised to include the additional charges on vehicle miles during congested periods.
- **High frequency transit:** 50% increase in frequency and transit revenue miles. The values in the two files, *marea_transit_service.csv* and *bzone_transit_service.csv*, were increased by 50% relative to the reference case.
- **Increased density:** 50% increase in mixed-use neighborhoods and intersection density. The values in the two files, *bzone_network_design.csv* and *bzone_urban-mixed-use_prop.csv*, were increased by 50% relative to the reference case.

The SKATS RSPM contains 232 Bzones and has a population of 273,913 persons in 104,569 households in the base year (2021). The population increases to 341,086 persons in 130,454 households in the future year (2050). This case study application is intended to test the technical functionality of prototype module, measure real-world model runtimes, and allow reasonableness checks of health impact estimates.

Case Study Results

The prototype model estimates yearly avoided mortality for several health outcomes: all-cause non-accidental mortality (i.e., all-cause mortality excluding accidents like traffic deaths), COPD mortality, diabetes mortality, colon cancer mortality, and traffic deaths. The same outcome measures are also expressed as yearly avoided years of life lost (YLL), which represent the future years of life lost due to excess mortality (i.e., a mortality event for a younger individual will have a higher YLL value than the same mortality event for an older individual, all else constant). In general, the prototype model produces reasonable results, with the highest health benefits generally estimated for high frequency transit scenarios (Table 1, Table 2). The model also runs quickly for these scenarios, with a total runtime of 60 seconds on a laptop with modest hardware.¹⁰

¹⁰ 12th Gen Intel(R) Core(TM) i7-1265U, 2700 Mhz, 10 Cores, 12 Logical Processors, and 16 GB RAM

Table 1. Prototype model mortality estimates

	SOV diversion	Congestion charge	High frequency transit	Increased density
Avoided all-cause mortality, excluding traffic deaths (deaths/year)	-0.14	0.39	2.95	0.27
Avoided COPD mortality (deaths/year)	-0.02	0.02	0.14	-0.02
Avoided diabetes mortality (deaths/year)	-0.01	0.02	0.15	0.01
Avoided colon cancer mortality (deaths/year)	0.00	0.00	0.02	0.00
Avoided traffic mortality (deaths/year)	0.02	0.17	0.22	-0.05

Table 2. Prototype model YLL estimates

	SOV diversion	Congestion charge	High frequency transit	Increased density
Avoided years of life lost, excluding traffic deaths (YLL/year)	-1.49	11.7	78.9	12.33
Avoided COPD years of life lost (YLL /year)	-0.26	0.40	2.15	-0.32
Avoided diabetes years of life lost (YLL /year)	-0.12	0.57	3.56	0.39
Avoided colon cancer years of life lost (YLL /year)	0.00	0.04	0.46	0.12
Avoided traffic years of life lost (YLL /year)	0.73	8.99	9.33	-2.09

Interestingly, the congestion charge and increased density scenarios tend to show similar health impacts. Conversely, the SOV diversion scenario tends to produce increased mortality and YLLs. This result highlights the utility of integrating a health impact model with a behavioral model that is sensitive to a wide range of policies. It should be noted that the health benefits of the transit scenario are higher than expected, but this may be because VE is using the ‘alt-models’ calculation of travel demand which is a relatively weak module within VE and while the increase in transit trips may be well captured, the resulting change in other travel modes (in this case, a substantial increase in walking) may not be accurate. This is an area for further consideration by VE users, but it is outside of the realms of the VE-ITHIM prototype.

Overall, this case study demonstration of the prototype model provides demonstrates: 1) the model is technically functional; 2) model runtime is acceptable; and 3) model outputs are generally within expected ranges.

Opportunities for Further Improvement

The prototype VE/ITHIM model integration on the sample RSPM results demonstrates a successful post-processor to VE models. There are opportunities to make future enhancements to the ITHIM scripts and its integration with VE. These enhancements include, but are not limited to:

- **Continue to make incremental improvements to code quality throughout the VE-ITHIM prototype.** ITHIM-R contains a mix of software development approaches (e.g., a mix of `data.table` and tidyverse functions are used to do similar things in different modules of the code). While substantial progress has been made on this front, numerous opportunities for further improvement exist.
- **Refine the traffic safety pathway technical approach.** Given the complexity of the approach to estimate traffic injuries in ITHIM-R, a simpler, rates-based approach was adopted from California ITHIM. However, neither of these approaches consider contextual factors (specifically, safety countermeasures and/or built environment influences on crash risks). Given the modularity of the prototype model, more advanced methods could be applied to this pathway in the future. The rates-based approach is like that used in the VE Travel Performance package using crash rates per mile of travel for various modes. The benefit is that those rates can change over time in the VE model providing some degree of flexibility to account for some of these countermeasures.
- **Complete other ITHIM-R disease pathways.** The prototype model contains 5 of the 21 disease pathways present in ITHIM-R. Future enhancements can increase the coverage of diseases present in ITHIM-R.
- **Automate input data preparation.** The Python scripts provided alongside the prototype model could be refined (and converted to R) so that input data are automatically generated if missing when the model is run.
- **Further refine parameter management.** The introduction of a standalone settings object represents a substantial improvement over settings management in ITHIM-R but lacks some best practices in settings management (e.g., validation of model inputs). However, this may be unnecessary if conversion of the prototype model to a full VE module is pursued.
- **Convert prototype model to a VE module.** The prototype model described here, while successful in meeting its established goals, exists as a standalone post-processor rather than a fully functional VE module. Further efforts to integrate this model into the VE framework and develop an ITHIM VE module could improve the usability of future iterations of the tool.